

Adaptive Quiz and Exam Engine

Presented by Group - 3
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Abstract

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The Adaptive Quiz & Exam Engine is a software application designed to conduct quizzes and examinations while adjusting question difficulty according to a student's performance. Traditional quiz systems usually present the same fixed set of questions to every student, which may not accurately reflect individual learning levels. This system provides a structured platform for managing questions, conducting assessments, recording answers, and adapting the difficulty of upcoming questions. The system allows an administrator or teacher to add, update, search, and organize questions using details such as question ID, subject, topic, difficulty level, options, and correct answer. During a quiz, the engine evaluates the student's responses and selects the next question based on the student's recent performance. Correct answers can lead to moderately harder questions, while incorrect answers can lead to easier questions. By automating question selection and score calculation, the proposed system reduces manual work, provides immediate feedback, and creates a more personalized assessment experience. The system can be implemented as a Java console application and can later be enhanced with a database, graphical interface, login system, question randomization, performance analytics, timers, and teacher dashboards.

Understanding Problem Statement

A university wants to develop an Adaptive Quiz and Exam Engine using Java to allow students to take subject-wise quizzes and instantly calculate their performance.

The system should first collect the student's name, roll number, and section. It should then allow the student to select one of three subjects:

PSP, MFC, DDCA

Each subject contains 10 multiple-choice questions, where each answer is represented by a number from 1 to 4. The system stores the correct answers for each subject in arrays.

After the student enters their answers, the system should:

- Compare each entered answer with the corresponding correct answer.
- Count the number of correct answers.
- Calculate the percentage based on 10 questions.
- Display the number of correct answers and the percentage.
- Display an error message if the student enters an invalid subject number.

After completing a quiz, the system should ask the student whether they want to continue with another subject. If the student enters **true**, another quiz can be attempted; if the student enters **false**, the program terminates.

CO-1 : Basic Java Programming Concepts

Java program structure

- Scanner
- Variables
- Data types
- Operators

CO-2: Control Flow and Decision Making

- while
- Switch
- case
- break
- default
- if
- for

CO-3: Methods and Modular Programming

- 1D Arrays
- Arrays traversal
- Counting
- Array processing

Geo Tag Photo and git hub link

<https://github.com/PVSrisuktaHiranmayi>